

Trigonometric ratios of $\frac{\theta}{2}$

We have

$$\begin{aligned}\cos 2\theta &= \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 \\ &= 1 - 2 \sin^2 \theta\end{aligned}$$

Replacing θ by $\frac{\theta}{2}$, we get

$$\begin{aligned}\cos \theta &= \cos^2 \theta/2 - \sin^2 \theta/2 \\ &= 2 \cos^2 \theta/2 - 1 = 1 - 2 \sin^2 \theta/2\end{aligned}$$

Similarly,

$$\sin 2\theta = 2 \sin \theta \cdot \cos \theta$$

$$\therefore \sin \theta = 2 \sin \theta/2 \cdot \cos \theta/2$$

Now,

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Replacing θ by $\frac{\theta}{2}$

$$\tan \theta = \frac{2 \tan \theta/2}{1 - \tan^2 \theta/2}$$

$$\begin{aligned} \text{Now, } \sin \theta &= 2 \sin \theta/2 \cdot \cos \theta/2 \\ &= \frac{2 \sin \theta/2 \cdot \cos \theta/2}{1} \end{aligned}$$

$$= \frac{2 \sin \theta/2 \cdot \cos \theta/2}{\sin^2 \theta/2 + \cos^2 \theta/2}$$

$$\sin^2 \theta/2 + \cos^2 \theta/2 = 1$$

Divide numerator and denominator by $\cos^2 \theta/2$

$$= \frac{\frac{2 \sin \theta/2 \cdot \cos \theta/2}{\cos^2 \theta/2}}{\frac{\sin^2 \theta/2 + \cos^2 \theta/2}{\cos^2 \theta/2}} = \frac{\frac{2 \sin \theta/2 \cdot \cos \theta/2}{\cos \theta/2 \cdot \cos \theta/2}}{\frac{\sin^2 \theta/2}{\cos^2 \theta/2} + \frac{\cos^2 \theta/2}{\cos^2 \theta/2}}$$

$$= \frac{\frac{2 \sin \theta/2}{\cos \theta/2}}{\frac{\sin^2 \theta/2}{\cos^2 \theta/2} + 1}$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta$$

$$\sin \theta = \frac{2 \tan \theta/2}{\tan^2 \theta/2 + 1}$$

$$\begin{aligned} \text{Now } \cos \theta &= \cos^2 \theta/2 - \sin^2 \theta/2 \\ &= \frac{\cos^2 \theta/2 - \sin^2 \theta/2}{1} \end{aligned}$$

$$= \frac{\cos^2 \theta/2 - \sin^2 \theta/2}{\cos^2 \theta/2 + \sin^2 \theta/2}$$

$$\cos^2 \theta/2 + \sin^2 \theta/2 = 1$$

Divide numerator and denominator by $\cos^2 \theta/2$

$$= \frac{\frac{\cos^2 \theta/2}{\cos^2 \theta/2} - \frac{\sin^2 \theta/2}{\cos^2 \theta/2}}{\frac{\cos^2 \theta/2}{\cos^2 \theta/2} + \frac{\sin^2 \theta/2}{\cos^2 \theta/2}}$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta$$

$$\cos \theta = \frac{1 - \tan^2 \theta/2}{1 + \tan^2 \theta/2}$$

Example 4.3.23 : If $\theta = 45^\circ$, find the value of $\cos \theta/2$



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Solution : We know that

$$1 + \cos \theta = 2 \cos^2 \theta/2$$

$$\therefore \cos^2 \theta/2 = \frac{1 + \cos \theta}{2}$$

$$\therefore \cos \theta/2 = \sqrt{\frac{1 + \cos \theta}{2}}$$

$$= \sqrt{\frac{1 + \cos 45^\circ}{2}}$$

$$\cos 45^\circ = 1/\sqrt{2}$$

$$= \sqrt{\frac{1 + 1/\sqrt{2}}{2}}$$

$$= \sqrt{\frac{\sqrt{2} + 1}{2\sqrt{2}}}$$

Example 4.3.26 : Prove that $\frac{\sin \theta}{1 + \cos \theta} = \tan \frac{\theta}{2}$.



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Solution : Consider

$$\text{L.H.S.} = \frac{\sin \theta}{1 + \cos \theta}$$

$$= \frac{2 \sin \theta/2 \cdot \cos \theta/2}{2 \cos^2 \theta/2}$$

$$= \frac{2 \sin \theta/2 \cdot \cos \theta/2}{2 \cos \theta/2 \cdot \cos \theta/2}$$

$$\frac{\sin \theta = 2 \sin \theta/2 \cdot \cos \theta/2}{1 + \cos \theta = 2 \cos^2 \theta/2}$$

$$= \frac{\sin \theta/2}{\cos \theta/2}$$

$$= \tan \theta/2$$

$$\therefore \frac{\sin \theta}{1 + \cos \theta} = \tan \theta/2$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta$$

Example 4.3.27 : Prove that $\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} = \tan \theta/2$.

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Solution : L.H.S. = $\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta}$

Rearranging the term

$$= \frac{1 - \cos \theta + \sin \theta}{1 + \cos \theta + \sin \theta}$$

$$\sin \theta = 2 \sin \theta/2 \cdot \cos \theta/2$$

$$1 - \cos \theta = 2 \sin^2 \theta/2$$

$$1 + \cos \theta = 2 \cos^2 \theta/2$$

$$= \frac{2 \sin^2 \theta/2 + 2 \sin \theta/2 \cdot \cos \theta/2}{2 \cos^2 \theta/2 + 2 \sin \theta/2 \cdot \cos \theta/2}$$

Taking $\sin \theta/2$ common in numerator and $\cos \theta/2$ in denominator

$$= \frac{2 \sin \theta/2 [\sin \theta/2 + \cos \theta/2]}{2 \cos \theta/2 [\cos \theta/2 + \sin \theta/2]}$$

$$= \frac{\sin \theta/2}{\cos \theta/2}$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta$$

$$= \tan \frac{\theta}{2} = \text{R.H.S.}$$

$$\therefore \frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} = \tan \frac{\theta}{2}$$

Example 4.3.29 : Prove that $\tan A + \sec A = \tan \left(\frac{\pi}{4} + \frac{A}{2} \right)$.

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Solution : We have

$$\text{R.H.S.} = \tan \left(\frac{\pi}{4} + \frac{A}{2} \right)$$

$$\tan (A + B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$$

$$= \frac{\tan \pi/4 + \tan A/2}{1 - \tan \pi/4 \cdot \tan A/2} = \frac{1 + \tan A/2}{1 - \tan A/2}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1 + \frac{\sin A/2}{\cos A/2}}{1 - \frac{\sin A/2}{\cos A/2}} = \frac{\frac{\cos A/2 + \sin A/2}{\cos A/2}}{\frac{\cos A/2 - \sin A/2}{\cos A/2}}$$

$$= \frac{\cos A/2 + \sin A/2}{\cos A/2 - \sin A/2}$$

$$= \frac{(\cos A/2 + \sin A/2)}{(\cos A/2 - \sin A/2)} \times \frac{(\cos A/2 + \sin A/2)}{(\cos A/2 + \sin A/2)}$$

$$= \frac{(\cos A/2 + \sin A/2)^2}{\cos^2 A/2 - \sin^2 A/2}$$

$$= \frac{(\cos^2 A/2 + 2 \sin A/2 \cdot \cos A/2 + \sin^2 A/2)}{\cos^2 A/2 - \sin^2 A/2}$$

$$\begin{aligned} \cos^2 A/2 + \sin^2 A/2 &= 1 \\ \sin A &= 2 \sin A/2 \cdot \cos A/2 \\ \cos A &= \cos^2 A/2 - \sin^2 A/2 \end{aligned}$$

$$= \frac{1 + \sin A}{\cos A} = \frac{1}{\cos A} + \frac{\sin A}{\cos A}$$

$$= \sec A + \tan A = \text{R.H.S.}$$

$$\tan A + \sec A = \tan \left(\frac{\pi}{4} + \frac{A}{2} \right)$$

Example 4.3.30 : Prove that $\frac{\cos A}{1 + \cos A} = \frac{1}{2} \left[1 - \tan^2 \frac{A}{2} \right]$

Solution : Consider

$$\text{L.H.S.} = \frac{\cos A}{1 + \cos A}$$

$$= \frac{\cos^2 A/2 - \sin^2 A/2}{2 \cos^2 A/2}$$

$$= \frac{1}{2} \left[\frac{\cos^2 A/2 - \sin^2 A/2}{\cos^2 A/2} \right]$$

$$= \frac{1}{2} \left[\frac{\cos^2 A/2}{\cos^2 A/2} - \frac{\sin^2 A/2}{\cos^2 A/2} \right]$$

$$= \frac{1}{2} [1 - \tan^2 A/2] = \text{R.H.S.}$$

$$\frac{\cos A}{1 + \cos A} = \frac{1}{2} \left[1 - \tan^2 \frac{A}{2} \right]$$

$$\begin{aligned} \cos \theta &= \cos^2 \theta/2 - \sin^2 \theta/2 \\ 1 + \cos \theta &= 2 \cos^2 \theta/2 \end{aligned}$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta$$